Dynamics Tutorial

29 March 2021 10:53

Question 1

- Find the inertia tensor of a right cylinder of homogenous density, with respect to a frame with origin at the center of mass of the body.

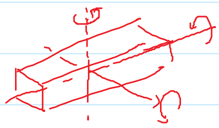
$F = ma$

force, mass, acceleration

$F = \text{big} \cdot \text{small}$

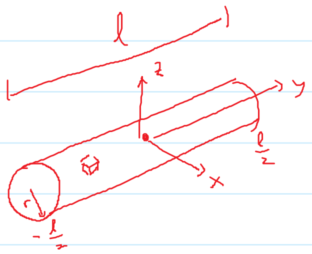
$F = \text{small} \cdot \text{big}$

resistance to motion



$N = I\alpha + \dots$

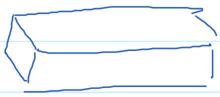
Moment, Inertia, rotational acceleration



$$\iiint_V \cdot \, \rho \, dV$$

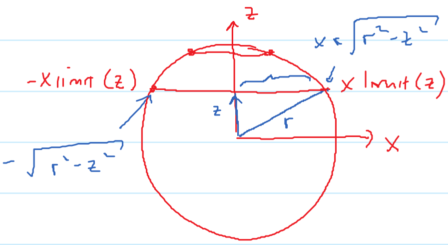
$$= \int_z \int_y \int_x \cdot \, \rho \, dx\, dy\, dz$$

Rectangular block



x, y, z independent of each other

But for cylinder, x & z are related !!!



For each level of z, the x-limits are different !!

How does x-limits vary wrt z ?

$$\Rightarrow \quad I_{xx} = \iiint_V (y^2+z^2)\,\rho\, \underbrace{dV}_{dx\,dy\,dz}$$

$$= \int_{-r}^{r} \int_{-\frac{l}{2}}^{\frac{l}{2}} \boxed{\int_{-\sqrt{r^2-z^2}}^{\sqrt{r^2-z^2}} (y^2+z^2)\,\rho \; dx} \, dy\, dz$$

$$= \int_{-r}^{r} \int_{-\frac{l}{2}}^{\frac{l}{2}} (y^2+z^2)\,x \Big|_{x=-\sqrt{r^2-z^2}}^{\sqrt{r^2-z^2}} \rho \; dy\, dz$$

$$= \int_{-r}^{r} \boxed{\int_{-\frac{l}{2}}^{\frac{l}{2}} 2(y^2+z^2)\sqrt{r^2-z^2}\;\rho \; dy} \, dz$$

$$= \int_{-r}^{r} 2\left(\frac{y^3}{3} + z^2 y\right)\sqrt{r^2-z^2}\;\rho \Big|_{y=-\frac{l}{2}}^{\frac{l}{2}} \, dz$$

$$= \int_{-r}^{r} 2\left(\frac{l^3}{24} + z^2\frac{l}{2} + \frac{l^3}{24} + z^2\frac{l}{2}\right)\sqrt{r^2-z^2}\;\rho \; dz$$

$$= \int_{-r}^{r} 2\left(\frac{l^3}{24} + z^2\frac{l}{2} + \frac{l^3}{24} + z^2\frac{l}{2}\right)\sqrt{r^2-z^2}\,\rho \quad dz$$

$$= 2\int_{-r}^{r} \left(\frac{l^3}{12} + z^2 l\right)\sqrt{r^2-z^2}\,\rho \quad dz$$

$$= 2\int_{-r}^{r} \frac{l^3}{12}\sqrt{r^2-z^2}\,\rho\, dz + 2\int_{-r}^{r} z^2 l\sqrt{r^2-z^2}\,\rho\, dz$$

refer to maths book for integration table

$$= 2\frac{l^3}{12}\rho\,\frac{1}{2}\left[\underbrace{z(r^2-z^2)}_{0} + r^2\underbrace{\arcsin\left(\frac{z}{r}\right)}_{\pi}\right]\Bigg|_{z=-r}^{r}$$

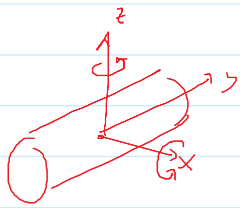
$$+ 2l\rho\left[-\frac{1}{4}\underbrace{z\sqrt{r^2-z^2}}_{0} + \frac{r^2}{8}\left[\underbrace{z\sqrt{r^2-z^2}}_{0} + r^2\underbrace{\arcsin\left(\frac{z}{r}\right)}_{\pi}\right]\right]\Bigg|_{z=-r}^{r}$$

here: $z(r^2-z^2)\big|_{z=-r}^{r} = 0$

$$\arcsin\left(\frac{z}{r}\right)\Big|_{z=-r}^{r} = \arcsin 1 - \arcsin(-1) = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$$

$$= \frac{l^3}{12}\rho\, r^2\pi + 2l\rho\frac{r^4}{8}\pi$$

$$= \left(\frac{l^2}{12} + \frac{r^2}{4}\right)\underbrace{\underbrace{\pi r^2 l}_{V}\,\rho}_{m}$$



$$\boxed{I_{xx} = \frac{1}{12}ml^2 + \frac{1}{4}mr^2}$$

$$\boxed{I_{zz} = \text{same as } I_{xx}}$$

$$I_{yy} = \iiint_V (x^2+z^2)\,\rho\, dV$$

$$= \int_{-r}^{r}\int_{-\frac{l}{2}}^{\frac{l}{2}}\boxed{\int_{-\sqrt{r^2-z^2}}^{\sqrt{r^2-z^2}} (x^2+z^2)\,\rho\, dx}\, dy\, dz$$

$$= \int_{-r}^{r}\int_{-\frac{l}{2}}^{\frac{l}{2}} \left(\frac{x^3}{3} + z^2 x\right)\rho\,\Bigg|_{-\sqrt{r^2-z^2}}^{\sqrt{r^2-z^2}} dy\, dz$$

$$= \int_{-r}^{r}\boxed{\int_{-\frac{l}{2}}^{\frac{l}{2}} \left(\frac{2\left(\sqrt{r^2-z^2}\right)^3}{3} + 2z^2\sqrt{r^2-z^2}\right)\rho\, dy}\, dz$$

$$= \int_{-r}^{r} \left(\frac{2\left(\sqrt{r^2-z^2}\right)^3}{3} + 2z^2\sqrt{r^2-z^2}\right) y\,\rho\,\Bigg|_{y=-\frac{l}{2}}^{\frac{l}{2}} dz$$

$$= \int_{-r}^{r} \left(\frac{2\left(\sqrt{r^2-z^2}\right)^3}{3} + 2z^2\sqrt{r^2-z^2}\right) l\,\rho\, dz$$

Integration table

$$= \left\{\frac{2}{3}r^2\frac{1}{2}\left[z\sqrt{r^2-z^2} + r^2\arcsin\left(\frac{z}{r}\right)\right] - \frac{4}{3}\cdot\frac{1}{4}\cdot z\sqrt{r^2-z^2} + \frac{4}{24}r^2\left[z\sqrt{r^2-z^2} + r^2\arcsin\left(\frac{z}{r}\right)\right]\right\}\Bigg|_{-r}^{r}\rho l$$

$z=-l$

$$= \frac{1}{2}\pi r^4 l \rho = \frac{1}{2} r^2 \underbrace{\underbrace{\pi r^2 l}_{V}\rho}_{M} = \frac{1}{2} m r^2$$

$$\boxed{I_{yy} = \frac{1}{2} m r^2}$$

$$I_{xy} = \iiint_V xy\,\rho\, dV$$

$$= \int_{-r}^{r}\int_{-\frac{l}{2}}^{\frac{l}{2}} \boxed{\int_{-\sqrt{r^2-z^2}}^{\sqrt{r^2-z^2}} xy\,\rho\, dx}\, dy\, dz$$

$$= \int_{-r}^{r}\int_{-\frac{l}{2}}^{\frac{l}{2}} \underbrace{\frac{1}{2}x^2 y\rho \Big|_{-\sqrt{r^2-z^2}}^{\sqrt{r^2-z^2}}}_{0}\, dy\, dz = \underline{\underline{0}}$$

Similarly, $I_{xz} = I_{yz} = \underline{\underline{0}}$

In summary,

$$\boxed{{}^{c}I = \begin{bmatrix} \frac{1}{12}ml^2 + \frac{1}{4}mr^2 & 0 & 0 \\ 0 & \frac{1}{2}mr^2 & 0 \\ 0 & 0 & \frac{1}{12}ml^2 + \frac{1}{4}mr^2 \end{bmatrix}}$$

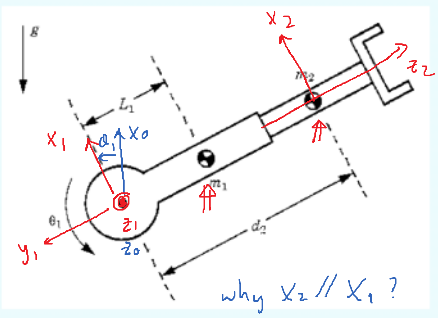
Question 2 RP

- Consider the following robot, with:

$${}^{c1}I_1 = \begin{bmatrix} I_{xx1} & 0 & 0 \\ 0 & I_{yy1} & 0 \\ 0 & 0 & I_{zz1} \end{bmatrix}$$

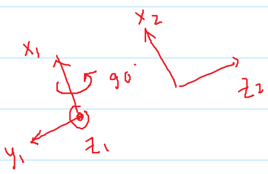
$${}^{c2}I_2 = \begin{bmatrix} I_{xx2} & 0 & 0 \\ 0 & I_{yy2} & 0 \\ 0 & 0 & I_{zz2} \end{bmatrix}$$

- Derive its dynamic equations.



why $x_2 \parallel x_1$?
because $\theta_n = 0$

1) Frames



2) Rotation matrices:

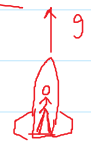
$${}^{0}_{1}R = \begin{bmatrix} c_1 & -s_1 & 0 \\ s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^{1}_{2}R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c90 & -s90 \\ 0 & s90 & c90 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

3) $\quad {}^1P_2 = \begin{bmatrix} 0 \\ -d_2 \\ 0 \end{bmatrix} \qquad {}^1P_{c1} = \begin{bmatrix} 0 \\ -L_1 \\ 0 \end{bmatrix} \qquad {}^2P_{c2} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

↑ Preparations

↓ Newton-Euler-Formula



Start with $\quad \underline{{}^0\omega_0 = 0}, \quad \underline{{}^0\dot{\omega}_0 = 0}, \quad {}^0\dot{v}_0 = \begin{bmatrix} g \\ 0 \\ 0 \end{bmatrix}$

Frame 1: $\quad {}^1\omega_1 = (\underbrace{{}^1_0R \cdot {}^0\omega_0}_{=0}) + (\dot{\theta}_1 \underbrace{{}^1\hat{z}_1}_{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}) = \underline{\begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}}$

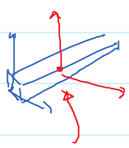
$${}^1\dot{\omega}_1 = ({}^1_0R \cdot \underbrace{{}^0\dot{\omega}_0}_{=0}) + ({}^1_0R \cdot \underbrace{{}^0\omega_0}_{=0} \times \dot{\theta}_1 \, {}^1\hat{z}_1) + \ddot{\theta}_1 \, {}^1\hat{z}_1 = \underline{\begin{bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{bmatrix}}$$

$${}^1\dot{v}_1 = ({}^1_0R \cdot {}^0\dot{v}_0) + (2 \, {}^1\omega_1 \times \underbrace{\dot{d}_1}_{=0 \text{ revolute}} \, {}^1\hat{z}_1) + (\underbrace{\ddot{d}_1}_{=0 \text{ revolute}} \, {}^1\hat{z}_1)$$

$$+ {}^1_0R \left[(\underbrace{{}^0\dot{\omega}_0}_{=0} \times {}^0P_1) + (\underbrace{{}^0\omega_0}_{=0} \times {}^0\omega_0 \times {}^0P_1) \right]$$

${}^0_1R^T$

$$= \begin{bmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} g \\ 0 \\ 0 \end{bmatrix} = \underline{\begin{bmatrix} c_1 g \\ -s_1 g \\ 0 \end{bmatrix}}$$



F = ma

COM$_1$: $\quad {}^1\dot{v}_{c1} = ({}^1\dot{v}_1) + ({}^1\dot{\omega}_1 \times {}^1P_{c1}) + ({}^1\omega_1 \times ({}^1\omega_1 \times {}^1P_{c1}))$

$$= \begin{bmatrix} c_1 g \\ -s_1 g \\ 0 \end{bmatrix} + \begin{vmatrix} i & j & k \\ 0 & 0 & \ddot{\theta}_1 \\ 0 & -L_1 & 0 \end{vmatrix} + \underline{\begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \times \begin{vmatrix} i & j & k \\ 0 & 0 & \dot{\theta}_1 \\ 0 & -L_1 & 0 \end{vmatrix}}$$

$$= \begin{bmatrix} c_1 g \\ -s_1 g \\ 0 \end{bmatrix} + \begin{bmatrix} L_1 \ddot{\theta}_1 \\ 0 \\ 0 \end{bmatrix} + \begin{vmatrix} i & j & k \\ 0 & 0 & \dot{\theta}_1 \\ L_1 \dot{\theta}_1 & 0 & 0 \end{vmatrix}$$

$$= \begin{bmatrix} c_1 g + L_1 \ddot{\theta}_1 \\ -S_1 g + L_1 \dot{\theta}_1^2 \\ 0 \end{bmatrix}$$

$F = ma \Rightarrow \quad {}^1F_1 = m_1 {}^1\dot{v}_{c_1} = \begin{bmatrix} m_1 c_1 g + m_1 L_1 \ddot{\theta}_1 \\ -m_1 S_1 g + m_1 L_1 \dot{\theta}_1^2 \\ 0 \end{bmatrix} \longrightarrow$ check: 2 dots $\ddot{\ }$ $\dot{\ }^2$ $\dot{\theta}_1 \dot{\theta}_2$

$${}^1N_1 = {}^{c_1}I_1 \cdot {}^1\dot{\omega}_1 + {}^1\omega_1 \times {}^{c_1}I_1 \, {}^1\omega_1$$

$$= \begin{bmatrix} I_{xx_1} & & \\ & I_{yy_1} & \\ & & I_{zz_1} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \times \begin{bmatrix} I_{xx_1} & & \\ & I_{yy_1} & \\ & & I_{zz_1} \end{bmatrix}}_{=0} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ I_{zz_1} \ddot{\theta}_1 \end{bmatrix}$$

1st link done ☺

Frame 2: $\quad {}^2\omega_2 = (\underbrace{{}^2_1R}_{{}^1_2R^T} \cdot {}^1\omega_1) + \underbrace{\dot{\theta}_2}_{=0} {}^2\hat{Z}_2$ prismatic

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ \dot{\theta}_1 \\ 0 \end{bmatrix}$$

$${}^2\dot{\omega}_2 = ({}^2_1R \cdot {}^1\dot{\omega}_1) + ({}^2_1R \cdot {}^1\omega_1 \times \underbrace{\dot{\theta}_2}_{=0 \text{ prismatic}} {}^2\hat{Z}_2) + (\underbrace{\ddot{\theta}_2}_{=0 \text{ prismatic}} {}^2\hat{Z}_2)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{bmatrix} = \begin{bmatrix} 0 \\ \ddot{\theta}_1 \\ 0 \end{bmatrix}$$

$${}^2\dot{v}_2 = ({}^2_1R \cdot {}^1\dot{v}_1) + (2\, {}^2\omega_2 \times \dot{d}_2 {}^2\hat{Z}_2) + (\ddot{d}_2 {}^2\hat{Z}_2)$$

$$+ {}^2_1R \left[({}^1\dot{\omega}_1 \times {}^1P_2) + {}^1\omega_1 \times ({}^1\omega_1 \times {}^1P_2) \right]$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} c_1 g \\ -S_1 g \\ 0 \end{bmatrix} + 2 \begin{vmatrix} i & j & k \\ 0 & \dot{\theta}_1 & 0 \\ 0 & 0 & \dot{d}_2 \end{vmatrix} + \begin{bmatrix} 0 \\ 0 \\ \ddot{d}_2 \end{bmatrix}$$

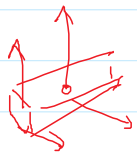
$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}\begin{bmatrix} -s_1 g \\ 0 \end{bmatrix} + 2\begin{vmatrix} 0 & \dot\theta_1 & 0 \\ 0 & 0 & \dot d_2 \end{vmatrix} + \begin{bmatrix} 0 \\ \ddot d_2 \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}\left\{ \begin{vmatrix} i & j & k \\ 0 & 0 & \ddot\theta_1 \\ 0 & -d_2 & 0 \end{vmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot\theta_1 \end{bmatrix} \times \begin{vmatrix} i & j & k \\ 0 & 0 & \dot\theta_1 \\ 0 & -d_2 & 0 \end{vmatrix} \right\}$$

$$= \begin{bmatrix} c_1 g \\ 0 \\ s_1 g \end{bmatrix} + 2\begin{bmatrix} \dot\theta_1 \dot d_2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \ddot d_2 \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}\left\{ \begin{bmatrix} d_2 \ddot\theta_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ d_2 \dot\theta_1^2 \\ 0 \end{bmatrix} \right\}$$

$$= \begin{bmatrix} c_1 g + 2\dot\theta_1 \dot d_2 + d_2 \ddot\theta_1 \\ 0 \\ s_1 g + \ddot d_2 - d_2 \dot\theta_1^2 \end{bmatrix}$$



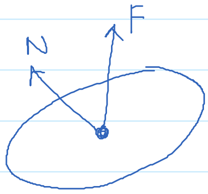
COM$_2$: $${}^2\dot v_{c2} = ({}^2\dot v_2) + ({}^2\dot\omega_2 \times \underbrace{{}^2P_{c2}}_{=0}) + ({}^2\omega_2 \times {}^2\omega_2 \times \underbrace{{}^2P_{c2}}_{=0})$$

$$= {}^2\dot v_2$$

$${}^2F_2 = m_2 \cdot {}^2\dot v_{c2} = \begin{bmatrix} m_2 c_1 g + 2m_2 \dot\theta_1 \dot d_2 + d_2 \ddot\theta_1 \\ 0 \\ m_2 s_1 g + m_2 \ddot d_2 - m_2 d_2 \dot\theta_1^2 \end{bmatrix}$$ ← check "2 dots"

$${}^2N_2 = {}^{C_2}I_2 \, {}^2\dot\omega_2 + {}^2\omega_2 \times {}^{C_2}I_2 \, {}^2\omega_2$$

$$= \begin{bmatrix} I_{xx2} & & \\ & I_{yy2} & \\ & & I_{zz2} \end{bmatrix}\begin{bmatrix} 0 \\ \ddot\theta_1 \\ 0 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ \dot\theta_1 \\ 0 \end{bmatrix} \times \begin{bmatrix} I_{xx2} & & \\ & I_{yy2} & \\ & & I_{zz2} \end{bmatrix}\begin{bmatrix} 0 \\ \dot\theta_1 \\ 0 \end{bmatrix}}_{=0}$$

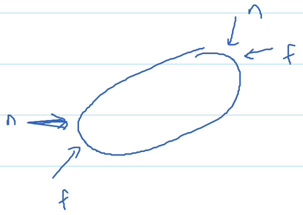


$$= \begin{bmatrix} 0 \\ I_{yy2} \ddot\theta_1 \\ 0 \end{bmatrix}$$

2^nd link done

Outward iteration done

inward iteration:



Start with $^3n_3 = 0$, $^3f_3 = 0$ (assume no contact with environment)

$$^2f_2 = {}^2_3R \cdot \underbrace{^3f_3}_{=0} + {}^2F_2 = \begin{bmatrix} ^2F_{2x} \\ 0 \\ ^2F_{2z} \end{bmatrix}$$

$$^2n_2 = {}^2_3R \underbrace{^3n_3}_{=0} + \underbrace{^2P_{c2}}_{=0} \times {}^2F_2 + {}^2P_3 \times {}^2_3R \cdot \underbrace{^3f_3}_{=0} + {}^2N_2 = \begin{bmatrix} 0 \\ ^2N_{2y} \\ 0 \end{bmatrix}$$

$$^1f_1 = {}^1_2R \cdot {}^2f_2 + {}^1F_1 = \text{not needed}$$

$$^1n_1 = {}^1_2R \cdot {}^2n_2 + {}^1P_{c1} \times {}^1F_1 + ({}^1P_2 \times {}^1_2R\, {}^2f_2) + {}^1N_1$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ ^2N_{2y} \\ 0 \end{bmatrix} + \begin{vmatrix} i & j & k \\ 0 & -L_1 & 0 \\ ^1F_{1x} & ^1F_{1y} & 0 \end{vmatrix}$$

$$+ \begin{bmatrix} 0 \\ -d_2 \\ 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} ^2F_{2x} \\ 0 \\ ^2F_{2z} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ ^1N_{1z} \end{bmatrix}$$

$$= \dots$$

$$= \begin{bmatrix} * \\ * \\ I_{yy2}\ddot{\theta}_1 + m_1 L_1 c_1 g + m_1 L_1^2 \ddot{\theta}_1 + m_2 c_1 d_2 g + m_2 d_2^2 \ddot{\theta}_1 + 2 m_2 d_2 \dot{\theta}_1 \dot{d}_2 + I_{zz1}\ddot{\theta}_1 \end{bmatrix}$$

Dynamics equations. $\tau_1 = 3^{rd}$ row of $^1n_1$ (revolute)

$\tau_2 = 3^{rd}$ row of $^2f_2$ (prismatic)

$$\boxed{\begin{aligned} \tau_1 &= I_{yy2}\ddot{\theta}_1 + m_1 L_1 c_1 g + m_1 L_1^2 \ddot{\theta}_1 + m_2 c_1 d_2 g + m_2 d_2^2 \ddot{\theta}_1 + 2 m_2 d_2 \dot{\theta}_1 \dot{d}_2 + I_{zz1}\ddot{\theta}_1 \\ \tau_2 &= {}^2F_{2z} = m_2 s_1 g + m_2 \ddot{d}_2 - m_2 d_2 \dot{\theta}_1^2 \end{aligned}}$$

↓ put this into $M(q)\ddot{q} + V(q,\dot{q}) + G(q) = \tau$ form.

$$\left[\begin{array}{c|c} I_{yy2} + m_1 L_1^2 + m_2 d_2^2 + I_{zz1} & 0 \\ \hline 0 & m_2 \end{array}\right] \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{d}_2 \end{bmatrix} + \begin{bmatrix} 2 m_2 d_2 \dot{\theta}_1 \dot{d}_2 \\ - m_2 d_2 \dot{\theta}_1^2 \end{bmatrix} + \begin{bmatrix} m_1 L_1 C_1 g + m_2 C_1 d_2 g \\ m_2 S_1 g \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}$$

M should be symmetrical